

Coral Poster Presentation Script

UPEX 2026 | Bullet-point script for quick review

How to use this file: Practice the full bullet script for a short presentation. Use the short version if you are tight on time.

Full Poster Script

Opening

- Good morning judges.
- My name is Muhammad Luqman Haziq Bin Muhamad Lofi.
- My project is Coral Reef Health Assessment via Convolutional Neural Network-Based Image Analysis.
- This project uses AI to classify coral reef images and explain the result using Grad-CAM heatmaps.

Problem Statement

- Coral monitoring is still mostly manual.
- It needs expert divers, field surveys, time, and cost.
- Coral bleaching is a serious threat caused by climate change and environmental stress.
- Many AI systems also have a black-box problem, where they give a result without explaining why.

Objectives

- Classify coral images into Healthy, Bleached, and Dead.
- Provide explainable AI using Grad-CAM.
- Evaluate performance using accuracy, precision, recall, F1-score, and confusion matrix.

Methodology

- Step 1: collect and organize coral images into Healthy, Bleached, and Dead classes.
- Step 2: split the dataset into training, validation, and test sets.
- Step 3: preprocess images by resizing to 224 x 224 and converting BGR to RGB.
- Step 4: apply augmentation such as flipping, rotation, zoom, shift, and brightness adjustment.
- Step 5: handle hard examples with oversampling and class weighting.
- Step 6: train an EfficientNetB0 ensemble with five random seeds.
- Step 7: apply SWA to improve generalization.
- Step 8: evaluate with standard metrics and Grad-CAM explainability.

Results

- The user uploads a coral image.
- The system predicts the class and shows the confidence score.
- Grad-CAM highlights the important area used by the model.
- ReefGuide support helps explain the result in simple language.
- The web application can classify Healthy, Bleached, and Dead coral images.
- Healthy coral is supported by natural coral texture.
- Bleached coral is supported by pale coral regions.
- Dead coral is supported by degraded coral areas.
- The system provides prediction, confidence, class probabilities, and Grad-CAM explanation for each uploaded image.

Commercialization & Impact

- The main product is a Coral Health AI Platform, a web-based solution for automated coral health assessment.
- The platform provides AI-powered analysis, explainable predictions, confidence scores, and scalable monitoring capabilities.
- Target users include marine agencies, environmental monitoring organizations, research institutions, and coral restoration teams.
- Key applications include coral health assessment, bleaching detection, and long-term reef monitoring.
- The system enables faster and more efficient coral monitoring compared to traditional manual methods.
- It helps organizations reduce monitoring costs and support data-driven decision-making.
- Overall, the project contributes to marine sustainability, biodiversity protection, and coral reef conservation efforts.

Closing

- This project provides an AI-based coral health assessment system that is predictive and explainable.
- It helps users classify coral condition, understand the AI decision, and support faster reef monitoring.
- Thank you.

Short Version If Time Is Limited

- Good morning judges.
- My project is Coral Reef Health Assessment via Convolutional Neural Network-Based Image Analysis.
- The problem is that coral monitoring is manual, expensive, and time-consuming, while many AI systems are difficult to trust because they work like a black box.
- My system classifies coral images into Healthy, Bleached, and Dead, and explains the prediction using Grad-CAM heatmaps.
- The pipeline covers data collection, splitting, preprocessing, augmentation, class balancing, EfficientNetB0 ensemble training, SWA optimization, final evaluation, and Grad-CAM explainability.
- In the web application, users upload a coral image, receive a prediction with confidence and class probabilities, and can view the Grad-CAM focus area.
- ReefGuide support helps explain the result in simple language.
- The platform can support marine agencies, environmental monitoring teams, research institutions, and coral restoration groups.
- Overall, this project provides a faster and more explainable AI tool for coral reef health monitoring.
- Thank you.

Poster Walkthrough Order

1. Start at the title and introduce the project.
2. Move to Problem Statement: manual monitoring, bleaching threat, AI black-box problem.
3. Move to Objectives: classify, explain, evaluate.
4. Move to Methodology: explain the 8-step pipeline.
5. Move to Results: show upload flow, prediction, confidence, Grad-CAM, and ReefGuide.
6. End at Commercialization & Impact: product, target users, use cases, and contribution.

Key Terms to Review

Term	Simple Meaning
CNN	A deep learning model used for image classification.
EfficientNetB0	The lightweight CNN backbone used for coral classification.
Grad-CAM	A heatmap method that shows which image regions influenced the prediction.
SWA	Stochastic Weight Averaging, used to improve model generalization.
Ensemble	Five models trained with different seeds and averaged for more stable prediction.
TTA	Test-Time Augmentation, where multiple image views are predicted and averaged.
Class Weighting	Higher penalty for difficult or minority classes during training.

Final Reminder

- Do not read too fast.
- Use simple words: classify, explain, evaluate, deploy.
- If judges ask about accuracy, mention that strong test performance still needs wider real-world validation.
- If judges ask about trust, point to Grad-CAM and confidence score.

- If judges ask about usefulness, explain marine agencies, monitoring teams, research institutions, and coral restoration groups.